

CHASSIS C1 – C8: MASTER SLOT TO DESTINATION JB REPORT

COMPLETE PLANT AUTOMATION I/O ROUTING DIRECTORY & FIELD DISTRIBUTION SCHEDULE

Project Ref: x2608003
Scope: Turnkey Automated CIP Skid System
Drawing Ref: KAL-SYS-DIA-C1C8-001

1. Master Chassis System Overview (C1 through C8)

Rack	Chassis Description & Functional Role	Enclosure Location	Slots	Total Pts	Active	Spare	DI	DO	AI	AO / BUS
C1	Main Controller Rack (1756-L950TPSXT)	Control Cabinet CA1 (MCR)	12	228	138	90	128	64	32	0
C2	Main Expansion Rack (High-Density I/O)	Control Cabinet CA1 (MCR)	11	286	203	83	157	96	32	0
C3	Remote I/O Rack 1 (Spray Dryer / Tower)	Field RIO Cabinet	9	109	70	39	16	32	52	8
C4	Remote I/O Rack 2 (Upper Spray Dryer)	Field RIO Cabinet	5	89	44	45	48	24	16	0
C5	Remote I/O Station 5 (Slurry Building)	Slurry Building 2nd Floor	5	113	57	56	64	32	16	0
C6	Motor Control Hardwired Interlocks	MCC Room Ground Floor	2	18	18	0	10	8	0	0
C7	MCC & Fieldbus Supervisory System	MCC Room Ground Floor	8	226	158	68	118	88	0	20
C8	Intrinsically Safe (IS) Field Barriers	Field Ex Barrier Enclosures	7	149	75	74	95	8	46	0
TOTAL	All 8 Chassis Combined	All Plant Enclosures	59	1218	763	455	636	352	194	28

2. Destination Junction Box Directory (16 Destinations)

Destination	Location	Process Functional Scope	Feeding Racks	Total Pts	Active	Spare
CA1	Main Control Room	Central PLC Automation & Network Cabinets	C1, C2, C3, C4, C5	40	0	40
IS-JB-603	Spray Dryer 3rd Floor (IS)	Intrinsically Safe Ex Zone Chamber Box	C8	32	22	10
IS-JB-608	Spray Dryer 8th Floor (IS)	Intrinsically Safe Ex Zone Top Level Box	C8	32	14	18
IS-JB-612	Packing Tower 2nd FI (IS)	Intrinsically Safe Ex Zone Lower Tower Box	C8	41	17	24
IS-JB-618	Packing Tower 8th FI (IS)	Intrinsically Safe Ex Zone Upper Tower Box	C8	44	22	22
JB-401	Infeed 2nd Floor	Infeed Raw Material Prep & Wet Mixing	C1	112	85	27
JB-402	Jet Cooker 2nd Floor	Jet Cooker Thermal Cooking & Starch Conversion	C2, C3	65	41	24
JB-601	Spray Dryer 1st Floor	Spray Dryer Bottom Discharge & Pumps	C1, C7	85	58	27
JB-602	Spray Dryer 3rd Floor	Spray Dryer Middle Body & Chamber Instruments	C2, C3	82	57	25
JB-606	Spray Dryer 6th Floor	Spray Dryer Air Filter & Observation Level	C4	26	7	19
JB-607	Spray Dryer 7th Floor	Spray Dryer Cyclone & Bag Filter Exhaust	C2, C3, C7	195	151	44
JB-608	Spray Dryer 8th Floor	Spray Dryer Air Dispenser & Scrubber Top	C4	6	1	5
JB-612	Packing Tower 2nd Floor	Packing Tower Product Discharge & Conveyor	C4, C7	57	37	20

JB-618	Packing Tower 8th Floor	Packing Tower Dehumidifier & Cyclone Air	C3	52	25	27
MCC	MCC Room Ground Floor	Main Motor Control Center & VFD Inverters	C6, C7	237	169	68
RIO-200	Slurry Out-Building 2nd Fl	Remote Slurry Processing Building Station	C5	112	57	55

3. Master Chassis C1–C8 Slot-to-Destination Schedule

Chassis	Slot	Module Card	Signal	Dest JB	Location	Total	Active	Spare	Sample Connected Loops / Tags
C1	00	1756-L950TPSXT	CPU	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C1	01	1756-EN4TR	ETH-1	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C1	02	1756-EN4TR	ETH-2	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C1	03	1756-EN4TR	ETH-3	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C1	04	1756-IB32	DI	CA1	Main Control Room	16	0	16	Reserve / Spare Bank
C1	04	1756-IB32	DI	JB-401	Infeed 2nd Floor	16	16	0	FSL-40201, FSL-40202, FT-40201 (+13 more)
C1	05	1756-IB32	DI	JB-401	Infeed 2nd Floor	32	32	0	XV-40201D, XV-40201D, XV-40201E (+29 more)
C1	06	1756-IB32	DI	JB-401	Infeed 2nd Floor	16	2	14	XV-40203E, XV-40203E
C1	06	1756-IB32	DI	JB-601	Spray Dryer 1st Floor	16	10	6	ES-60202, FS-60205, LSL-60201 (+7 more)
C1	07	1756-IB32	DI	JB-601	Spray Dryer 1st Floor	32	23	9	XA-60202A, XA-60202B, XA-60202C (+20 more)
C1	08	1756-OB32	DO	CA1	Main Control Room	16	0	16	Reserve / Spare Bank
C1	08	1756-OB32	DO	JB-401	Infeed 2nd Floor	16	16	0	XV-40202E, XV-40202F, XV-40202G (+13 more)
C1	09	1756-OB32	DO	JB-401	Infeed 2nd Floor	16	8	8	XV-40201E, XV-40201F, XV-40201G (+5 more)
C1	09	1756-OB32	DO	JB-601	Spray Dryer 1st Floor	16	12	4	SV-60201, SV-60202, SV-60230 (+9 more)
C1	10	1756-IF16	AI	JB-401	Infeed 2nd Floor	16	11	5	FT-40201, FT-40201, FT-40202 (+8 more)
C1	11	1756-IF16	AI	JB-601	Spray Dryer 1st Floor	16	8	8	DPI-60240, DPT-60202, FT-60201 (+5 more)
C2	00	1756-EN4TR	ETH-1	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C2	01	1756-IB32	DI	JB-402	Jet Cooker 2nd Floor	32	24	8	FS-40204, FS-40205, FT-40221 (+21 more)
C2	02	1756-IB32	DI	JB-602	Spray Dryer 3rd Floor	32	23	9	FT-60202, LSH-60251, PSL-60220 (+20 more)
C2	03	1756-IB32	DI	JB-607	Spray Dryer 7th Floor	29	29	0	XA-FH-60210A, XA-FH-60210B, XA-FH-60210C (+26 more)
C2	04	1756-IB32	DI	JB-607	Spray Dryer 7th Floor	32	32	0	XV-60261FC, XV-60261FC, XV-60264SQ (+29 more)
C2	05	1756-IB32	DI	JB-607	Spray Dryer 7th Floor	32	20	12	XV-60262SH, XV-60262SH, XV-60263 (+17 more)
C2	06	1756-OB32	DO	JB-402	Jet Cooker 2nd Floor	16	9	7	XV-40203F, XV-40204B, XV-40204C (+6 more)
C2	06	1756-OB32	DO	JB-602	Spray Dryer 3rd Floor	16	3	13	XS-60240B, XV-60214, XV-60241
C2	07	1756-OB32	DO	JB-607	Spray Dryer 7th Floor	32	31	1	SV-60207, XS-FH-60210C, XS-FH-60210D (+28 more)
C2	08	1756-OB32	DO	JB-607	Spray Dryer 7th Floor	32	9	23	XV-60264, XV-60264FN, XV-60264FP (+6 more)
C2	09	1756-IF16	AI	JB-402	Jet Cooker 2nd Floor	16	7	9	FT-40221, FT-40221, FT-40222 (+4 more)
C2	10	1756-IF16	AI	JB-602	Spray Dryer 3rd Floor	16	16	0	AT-60241, DPT-60203, FT-60202 (+13 more)
C3	00	1756-EN4TR	ETH-1	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C3	01	1756-IB32	DI	JB-618	Packing Tower 8th Floor	11	7	4	DV-61301, DV-61301, DV-61302 (+4 more)
C3	02	1756-IB32	DI	JB-618	Packing Tower 8th Floor	5	5	0	PSL-61311, SV-61304, SV-61304 (+2 more)
C3	03	1756-OB32	DO	JB-618	Packing Tower 8th Floor	32	9	23	DV-61301, DV-61302, SV-61301 (+6 more)
C3	05	1756-IF16	AI	JB-618	Packing Tower 8th Floor	4	4	0	PT-61303, PT-61306, TT-61303 (+1 more)

C3	06	1756-N2	AI	JB-602	Spray Dryer 3rd Floor	16	13	3	TT-60215B, TT-60216, TT-60240 (+10 more)
C3	07	1756-N2	AI	JB-607	Spray Dryer 7th Floor	16	16	0	FT-60207, FT-60221, FT-60261FA (+13 more)
C3	08	1756-N2	AI	JB-607	Spray Dryer 7th Floor	16	11	5	FT-60264FQ, FT-60264FR, PT-60201 (+8 more)
C3	11	1756-N2	AO	JB-402	Jet Cooker 2nd Floor	1	1	0	TCV-40222
C3	11	1756-N2	AO	JB-602	Spray Dryer 3rd Floor	2	2	0	TV-60202, TV-60214
C3	11	1756-N2	AO	JB-607	Spray Dryer 7th Floor	5	2	3	PV-60201, UY-FH-60210B
C4	00	1756-EN4TR	ETH-1	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C4	01	1756-IB32	DI	JB-606	Spray Dryer 6th Floor	16	4	12	FT-60203, LSH-60250, XA-60261 (+1 more)
C4	02	1756-IB32	DI	JB-612	Packing Tower 2nd Floor	32	17	15	PSL-61305, PSL-61306, PVM-61301 (+14 more)
C4	04	1756-OB32	DO	JB-606	Spray Dryer 6th Floor	8	1	7	SV-60220
C4	04	1756-OB32	DO	JB-612	Packing Tower 2nd Floor	16	15	1	PVM-61301, PVM-61301, SV-61307 (+12 more)
C4	05	1756-IF16	AI	JB-606	Spray Dryer 6th Floor	2	2	0	DPT-60201, FT-60203
C4	05	1756-IF16	AI	JB-608	Spray Dryer 8th Floor	6	1	5	TT-60245
C4	05	1756-IF16	AI	JB-612	Packing Tower 2nd Floor	8	4	4	PT-61302, PT-61309, PT-61302 (+1 more)
C5	00	1756-EN4TR	ETH-1	CA1	Main Control Room	1	0	1	Reserve / Spare Bank
C5	01	1756-IB32	DI	RIO-200	Slurry Out-Building 2nd Fl	32	32	0	FS-20201, FT-913201, FT-936201 (+29 more)
C5	02	1756-IB32	DI	RIO-200	Slurry Out-Building 2nd Fl	32	8	24	XV-20211, XV-20211, XV-20212 (+5 more)
C5	03	1756-OB32	DO	RIO-200	Slurry Out-Building 2nd Fl	32	8	24	XA-92201, XV-20201, XV-20202 (+5 more)
C5	04	1756-IF16	AI	RIO-200	Slurry Out-Building 2nd Fl	16	9	7	EXT-BIILER, FT-913201, FT-913201 (+6 more)
C6	05	DO	DO	MCC	MCC Room Ground Floor	8	8	0	PCM-2030, PCM-2030, PCM-2031 (+5 more)
C6	06	DI	DI	MCC	MCC Room Ground Floor	10	10	0	PCM-2030, PCM-2030, PCM-2031 (+7 more)
C7	00	BUS	BUS	JB-601	Spray Dryer 1st Floor	1	1	0	PCY-60205
C7	00	BUS	BUS	JB-607	Spray Dryer 7th Floor	1	1	0	COM-FH-60210B
C7	00	BUS	BUS	JB-612	Packing Tower 2nd Floor	1	1	0	Communication Signal
C7	00	BUS	BUS	MCC	MCC Room Ground Floor	17	17	0	ATY-40201, ATY-40202, ATY-40203 (+14 more)
C7	01	DI	DI	MCC	MCC Room Ground Floor	32	24	8	ATY-40201, ATY-40201, ATY-40201 (+21 more)
C7	02	DI	DI	JB-601	Spray Dryer 1st Floor	2	2	0	PCY-60205, PCY-60205
C7	02	DI	DI	MCC	MCC Room Ground Floor	30	30	0	PCM-60205-4, PCM-60205-5, PCM-60205-5 (+27 more)
C7	03	DI	DI	JB-601	Spray Dryer 1st Floor	1	1	0	PCY-60205
C7	03	DI	DI	MCC	MCC Room Ground Floor	25	25	0	RVM-61302, RVM-61302, RVM-61303 (+22 more)
C7	04	DI	DI	MCC	MCC Room Ground Floor	28	0	28	Reserve / Spare Bank
C7	05	DO	DO	MCC	MCC Room Ground Floor	32	24	8	ATY-40201, ATY-40202, ATY-40203 (+21 more)
C7	06	DO	DO	JB-601	Spray Dryer 1st Floor	1	1	0	PCY-60205
C7	06	DO	DO	MCC	MCC Room Ground Floor	31	31	0	BFY-60240, PCY-60215, XS-60240A (+28 more)
C7	07	DO	DO	MCC	MCC Room Ground Floor	24	0	24	Reserve / Spare Bank
C8	01	DI	DI	IS-JB-603	Spray Dryer 3rd Floor (IS)	16	15	1	AT-40201, AT-40202, AT-40203 (+12 more)

C8	01	DI	DI	IS-JB-608	Spray Dryer 8th Floor (IS)	16	7	9	ZS-60201A, ZS-60201B, ZS-60231C (+4 more)
C8	02	DI	DI	IS-JB-612	Packing Tower 2nd FI (IS)	31	15	16	ISV-61303, ISV-61304, LSH-61304 (+12 more)
C8	03	DI	DI	IS-JB-618	Packing Tower 8th FI (IS)	32	13	19	ISV-61301, LSH-61301, LSH-61303 (+10 more)
C8	04	DO	DO	IS-JB-608	Spray Dryer 8th Floor (IS)	8	1	7	XA-60201
C8	05	AI	AI	IS-JB-603	Spray Dryer 3rd Floor (IS)	16	7	9	PT-60230, PT-60231, TT-60224 (+4 more)
C8	06	AI	AI	IS-JB-608	Spray Dryer 8th Floor (IS)	8	6	2	PT-60220, TT-60220, TT-60221 (+3 more)
C8	06	AI	AI	IS-JB-612	Packing Tower 2nd FI (IS)	10	2	8	DPT-61310, DPT-61312
C8	07	AI	AI	IS-JB-618	Packing Tower 8th FI (IS)	12	9	3	DPT-61301, DPT-61302, LT-61301 (+6 more)